



LESSON 4.2

FINDING THE AREA OF TRAPEZOIDS, PARALLELOGRAMS, AND RHOMBI

LESSON INTRODUCTION

MA.7.GR.1.1 Apply formulas to find the areas of trapezoids, parallelograms, and rhombi.

LEARNING TARGETS

- I can find the area of trapezoids, parallelograms, and rhombi.

BENCHMARK COHERENCE

BUILDING ON

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

WORKING TOWARD

MA.912.GR.4.4 Solve mathematical and real-world problems involving the area of two-dimensional figures.

ESSENTIAL QUESTIONS

- How do you find the areas of trapezoids, parallelograms, and rhombi in mathematical and real-world problems?

LESSON NARRATIVE

Students find the areas of trapezoids, parallelograms, and rhombi using the formulas they developed in Lesson 1. The lesson prepares students for lessons 3 and 4, where they will find the areas of composite figures.

COMPONENT SUMMARY

4.2.1 WARM-UP (~10 MIN.)

Watch and reflect on video on theme park engineer career

4.2.3 YOUR TURN! (20 MIN.)

Solve real-world area problems

4.2.2 GUIDED PRACTICE (15 MIN.)

Recall and apply area formulas

4.2.4 WRAP-UP (~5 MIN.)

Identify, correct errors, and find area of rhombi

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Parallelogram
- Trapezoid
- Rhombus

MATERIALS

- (Optional) Graph paper
- (Optional) Pattern blocks
- (Optional) Chart paper & markers

ADDITIONAL PREPARATION

- Warm-Up 4.2.1 contains a video to accompany the question prompts.
- Consider using data and/or observations from Lesson 1 and this lesson to determine small group intervention that can occur during 4.2.3 Your Turn!

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may use the diagonal length instead of the perpendicular length for the height.
- Students may neglect to use the order of operations when calculating the areas of trapezoids using the formula $A = \frac{1}{2}h(b_1 + b_2)$.
- Students may not include the units with their answers or may not raise the units to the power of 2.

THINGS TO CONSIDER

- Create and display an Anchor Chart with formulas, examples, and images. Show students how understanding the formulas prevents the need to memorize them. Include the various forms of the formulas that students derived in Lesson 1.
- Consider allowing students to use calculators so they can focus on applying the formulas.
- Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers. For example, question 2 in 4.2.2 could be a no calculator problem but the questions in 4.2.3 could be calculator problems.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

4.2.2 Guided Practice provides an opportunity for students to think about and discuss the similarities and differences in the area formulas. Students also discuss how the formulas connect to the physical characteristics of the shapes.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students apply the area formulas for trapezoids, parallelograms, and rhombi to solve real-world problems in 4.2.3 Your Turn! Student interest and motivation will be ignited as they see the utility of the area formulas.

DIFFERENTIATE INSTRUCTION

Consider providing students with graph paper or pattern blocks so they can physically model the real-world parallelograms, trapezoids, and rhombi in 4.2.3 Your Turn!

Alternatively, challenge students to a “Rhombus Scavenger Hunt” in which they must find and document as many different rhombi as possible as homework or an extension project if including dimensions and area.

4.2.1 WARM-UP: A DAY IN THE LIFE

QUESTIONING

Consider inviting a guest speaker, such as the technology teacher, an engineer, designer, or architect to join the class who can answer students' questions about their careers and the mathematics they use.



4.2.1 Warm-Up: A Day in the Life

Betsy McIver, a mechanical engineer, works in the Walt Disney Imagineering area and designs systems for the interactive attractions at the theme park. She mentions that her job can include creative left-brain thinking, as well as technical right-brain thinking.



She says that to be successful at her job, it is important to understand math, design, and theatre. She also mentions that many disciplines, such as designers and architects, all work together to make everything possible in the entertainment engineering industry.

1. Describe a ride you saw that looked exciting and entertaining.

Answers will vary. I saw a ride that went through Monsters Inc. that looked exciting.



4.2.2 Guided Practice: Finding the Area of Parallelograms, Trapezoids, and Rhombi

- Complete the table to match each type of quadrilateral with the formula to find its area.

$A = \frac{1}{2}h(b_1 + b_2)$
 $A = \frac{1}{2}d_1d_2$
 $A = bh$

Quadrilateral	Formula for the Area
parallelogram	$A = bh$
trapezoid	$A = \frac{1}{2}h(b_1 + b_2)$
rhombus	$A = \frac{1}{2}d_1d_2$

- The dimensions of three quadrilaterals are given in the table. Label each figure shown with the given dimensions in centimeters (cm). Then, calculate the area of each figure using the appropriate formula. Show your work.

Figure	Dimensions	Area
Parallelogram 	$b = 22\text{ cm}$ $h = 8\text{ cm}$	$A = bh$ $A = 22 \cdot 8$ $A = 176\text{ cm}^2$

Figure	Dimensions	Area
Trapezoid 	$b_1 = \frac{7}{12}\text{ cm}$ $b_2 = \frac{11}{12}\text{ cm}$ $h = \frac{1}{4}\text{ cm}$	$A = \frac{1}{2}h(b_1 + b_2)$ $A = \frac{1}{2} \cdot \frac{1}{4} \cdot \left(\frac{7}{12} + \frac{11}{12}\right)$ $A = \frac{3}{16}\text{ cm}^2$
Rhombus 	$d_1 = 6.5\text{ cm}$ $d_2 = 12.25\text{ cm}$	$A = \frac{1}{2}d_1d_2$ $A = \frac{1}{2} \cdot 6.5 \cdot 12.25$ $A = 39.8125\text{ cm}^2$

4.2.2 GUIDED PRACTICE: FINDING THE AREA OF PARALLELOGRAMS, TRAPEZOIDS, AND RHOMBI

TEACHER MOVES

Compare and Connect

Provide students with an opportunity to think about and discuss the similarities and differences in the area formulas. Students can also discuss how the formulas connect to the physical characteristics of the shapes.

Allow students to add equivalent formulas from Lesson 1 that they prefer to use.

TEACHER MOVES

The orientation of figures may be difficult for some students. Help students experience figures drawn with different orientations by purposely drawing figures differently.

4.2.3 YOUR TURN!

TEACHER MOVES

Consider pulling a small group during this activity to provide additional support to students based on observations from Lesson 1 and Lesson 2.

ENRICHMENT

Does it matter which diagonal length you identify as d_1 and which you identify as d_2 ?

ENRICHMENT

Why do you use square units when writing the area of rhombi, parallelograms, and trapezoids?

ENRICHMENT

In problems 3 and 5, does it matter which length you identify as the base and which you identify as the height?

ENRICHMENT

Does it matter which base length you identify as b_1 and which you identify as b_2 ? Justify your response using numbers.

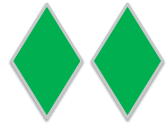
ENRICHMENT

What does the question mean by “face” of the building? What shape is that?



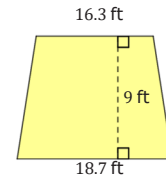
4.2.3 Your Turn!

- Alina is making rhombus-shaped earrings with green stones. The measure of one diagonal of each earring is $1\frac{3}{8}$ centimeters (cm), and the length of the other diagonal measures $1\frac{4}{9}$ cm. What is the area, in square centimeters (cm^2), of two earrings?



$$1\frac{71}{72}\text{cm}^2$$

- One wall of Kel's apartment is shaped like a trapezoid with one base 18.7 feet (ft), the other base 16.3 ft, and a height of 9 ft.

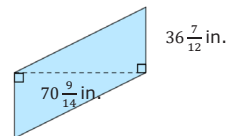


Kel plans to paint this wall blue and needs to know how much paint to buy. What is the area of the wall Kel plans to paint?

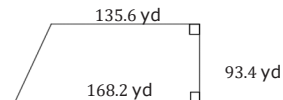
$$157.5\text{ft}^2$$

- Mika, a designer, is creating a geometric design in a tile floor. The design will be in the shape of a parallelogram with $36\frac{7}{12}$ inch (in.) bases and a height of $70\frac{9}{14}$ in. How many square inches of tile will she need for the design?

$$2,584\frac{59}{168}\text{in}^2$$



- The Smart family's horse pasture is in the shape of a trapezoid. One of the parallel sides is 168.2 yards (yd) long and the other is 135.6 yd long. The distance between these sides is 93.4 yd.



What is the area of the horse pasture?

$$14,187.46\text{yd}^2$$

- The Dockland Building in Hamburg, Germany is shaped like a parallelogram.

If the base of the building is 86 meters (m) and its height is 47 m, what is the area of the face of the building shown in the image?

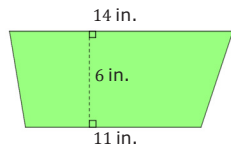


$$4,042\text{m}^2$$



4.2.4 Wrap-Up: Ticket Out the Door

1. Don's work for finding the area of the trapezoid is shown below.



Don's Work

$$\begin{aligned}
 A &= \frac{1}{2} h(b_1 + b_2) \\
 A &= \frac{1}{2} (14)(11 + 6) \\
 A &= \frac{1}{2} (14)(17) \\
 A &= \frac{1}{2} \cdot 238 \\
 A &= 119 \text{ in}^2
 \end{aligned}$$

- a. Write Don a note explaining what he did correctly and where he made an error.

Answers will vary. Don used the correct formula but he substituted the incorrect values for the base and the height and found the incorrect area.

- b. What is the area of the trapezoid?

75 in²

4.2.4 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

Students who struggle to express themselves in writing can circle the errors and then correct them or use an audio recorder (i.e. Flip Grid) or speech-to-text software.